Supplemental Material for "Data-Side Robustness of Training-Data Influence Estimation"

This appendix presents additional experiments and analyses that complement the results in the main paper. Appendix 1 provides additional experimental details, including the validation of the proposed aggregation strategy, runtime evaluation, and checkpoint budget analysis. Appendix 2 presents additional results for the training set size experiments on a real-world dataset. Appendix 3 provides further analyses of feature dimensionality, including distributional sanity checks and zero-padding control experiments. Appendix 4 reports additional analyses of high-loss samples, including loss-decile and conditional correlation analyses. Appendix 5 extends the neighbourhood and local-density experiments to the Diamonds dataset. Appendix 6 presents additional results for the class-imbalance experiments, including class-specific analyses and loss-adjusted regression results. Finally, Appendix 7 reports the complete downstream noisy-label detection and pruning results across all experimental settings. Together, these additional experiments provide further evidence for the robustness and generality of the observations reported in the main paper.

1 ADDITIONAL EXPERIMENTAL DETAILS

1.1 Aggregation Strategy

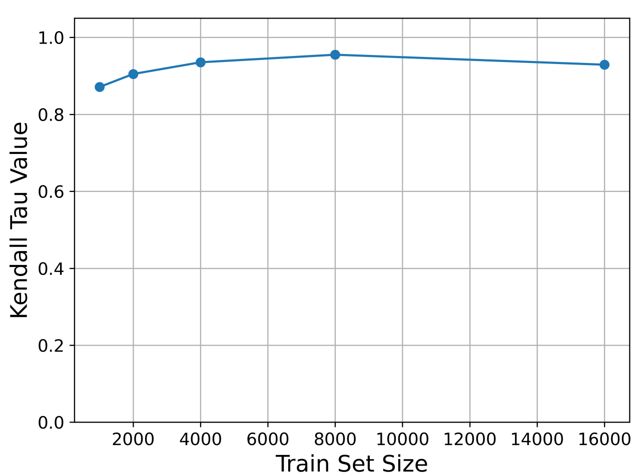


Figure 1: Kendall Tau correlation between the influence rankings produced by the highest-loss test sample used in prior work [1] and the proposed median-based aggregation across different training set sizes.

Our choice of median aggregation is motivated by both empirical validation and methodological consistency with prior work. Previous studies on influence fragility [1] evaluate stability using influence values computed on selected representative test points, typically the highest-loss sample. While this provides a useful measure of stability, it reflects only the behaviour of one test instance and therefore offers a local view of influence estimation.

In contrast, we aggregate influence scores across all test instances using the median, providing a dataset-level summary that is less sensitive to individual test samples or extreme influence values. To verify that this aggregation preserves the behaviour reported in prior work, we compare the rankings produced by our median-based aggregation with those obtained using the highest-loss test sample across different training set sizes.

Figure 1 shows that the resulting Kendall Tau values remain consistently high, ranging from approximately 0.88 to 0.95 across all training set sizes. This strong agreement indicates that the median aggregation largely preserves the influence ranking produced by the representative highest-loss test sample while providing a more robust dataset-level measure. The consistently high correlation demonstrates that the median-based aggregation captures the same underlying stability behaviour reported in [1], supporting its use throughout our experiments.

Therefore, the proposed aggregation method is not an arbitrary design choice but a principled extension of the evaluation protocol adopted in prior work, enabling influence stability to be analysed from a global rather than instance-specific perspective.

1.2 Runtime Evaluation

Table 1 reports the computational cost of FOIF and TracIn under different dataset sizes and feature dimensions. Each configuration was executed three times, and the reported runtimes are presented as the mean ± standard deviation across the three runs. Since all experiments were conducted without GPU acceleration, the reported runtimes represent conservative estimates and are expected to be substantially lower when reproduced using modern GPU hardware. It is worth noting that TracIn was evaluated using the full set of training checkpoints. Consequently, part of the additional computational cost compared with FOIF arises from accumulating influence over all checkpoints.

Table 1: Average runtime (mean ± standard deviation, in seconds) of the FO Influence Function (FOIF) and TracIn (TC) under different dataset sizes and feature dimensions over three independent runs.

# Samples	# Features	FOIF (s)	TC (s)
1,000	10	18.91 ± 5.11	97.36 ± 3.59
10,000	10	354.33 ± 36.34	494.83 ± 43.71
1,000	52	45.93 ± 0.26	141.62 ± 35.64
10,000	52	629.57 ± 2.46	702.91 ± 35.80

1.3 Effect of Checkpoint Budget

Throughout this paper, we use the full set of 300 checkpoints for TracIn to ensure that all experiments are based on the complete algorithm rather than an approximation. Table 2 evaluates the trade-off between computational cost and ranking quality when using fewer checkpoints. The results show that reduced checkpoint budgets substantially decrease runtime while maintaining highly similar influence rankings, suggesting that they provide a practical alternative when computational efficiency is a priority.

2 ADDITIONAL RESULTS FOR TRAINING SET SIZE

2.1 Number of Samples on Real Dataset - Diamonds

The Diamonds experiment on number of samples

The experimental setting follows the same protocol as the synthetic dataset experiments. To obtain a subset of 10,000 training samples and 500 test samples that approximately share the same underlying distribution, we first randomly selected a candidate centre point and retrieved its 10,500 nearest neighbours. Candidate centres were repeatedly sampled until the resulting neighbourhood exhibited a balanced label distribution. We then verified that the selected neighbourhood was sufficiently compact. The final neighbourhood had a mean distance of 1.80 from the centre point and a maximum distance of 2.23, compared with approximately 3.10 and 23.10, respectively, for a randomly selected subset of the same size.

The selected 10,500 samples were then randomly partitioned into ten chunks. The average distance of each chunk to the centre point ranged from 1.80 to 1.82, indicating that all chunks were drawn from a similar local distribution. This setup closely mimics the synthetic experiments, where all samples originate from the same Gaussian cluster, while preserving the characteristics of a real-world dataset. The resulting chunk-contribution entropy is reported in Table 3. Due to the computational cost of the real-data experiments, each configuration was repeated three times.

3 ADDITIONAL RESULTS FOR FEATURE DIMENSIONALITY

Huiping: soemthing maybe useful and we can do when we have time, varying Gaussian noise $\sigma \in \{0, 0.2, 0.5, 1\}$, this will definatly goes to appendix

3.1 Distributional Sanity Check

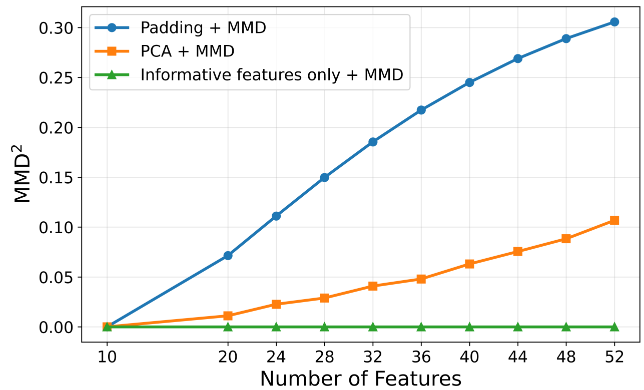


Figure 2: Maximum Mean Discrepancy (MMD) measured under three different dataset representations as the number of features increases. Smaller MMD values indicate greater distributional similarity.

To verify whether the degradation in influence stability is caused solely by increasing feature dimensionality or by unintended changes in the underlying dataset distribution, we measure the similarity between datasets using the Maximum Mean Discrepancy (MMD) [2]. MMD is a kernel-based statistical measure that quantifies the discrepancy between two empirical distributions by comparing the average pairwise similarities of samples within each dataset and across the two datasets. Intuitively, MMD compares how similar samples are within each dataset against how similar samples are across the two datasets. Formally, the squared MMD is defined as

$$\begin{aligned}\mathrm{MMD}^2(X,Y) = \mathbb{E}_{x,x'\sim X}[k(x,x')] + \mathbb{E}_{y,y'\sim Y}[k(y,y')] \\ - 2\mathbb{E}_{x\sim X,\ y\sim Y}[k(x,y)].\end{aligned} \tag{1}$$

where $k(\cdot,\cdot)$ denotes a kernel function. The first two terms measure the average similarity between samples from the same dataset, whereas the third term measures the average similarity between samples from different datasets. Consequently, smaller MMD values indicate that two datasets have more similar distributions, while larger values indicate greater distributional differences. In this work, we employ the Gaussian Radial Basis Function (RBF) kernel

$$k(x,y) = \exp(-\gamma\|x-y\|^2), \tag{2}$$

where γ is the kernel bandwidth parameter that controls how rapidly similarity decreases with increasing Euclidean distance. Following common practice, we estimate γ using the median heuristic from the baseline 10-feature dataset and keep it fixed across all MMD comparisons to ensure consistency across different feature dimensions.

Since kernel-based similarity measures compare samples in a common feature space, the datasets must share the same feature dimensionality. However, the datasets considered in our feature-dimensionality experiment contain different numbers of features. We therefore evaluate MMD under three complementary representations, each capturing a different aspect of dataset similarity.

Table 2: Effectiveness and runtime of TracIn under different checkpoint budgets. The influence rankings obtained using reduced checkpoint sets are compared against the full 300-checkpoint TracIn ranking using Weighted Kendall Tau and Top-10% overlap. Runtime corresponds to the TracIn influence estimation stage only.

Checkpoints	Runtime	Weighted Kendall Tau	Top-10% Overlap
5%	148.26	0.9405	0.7887
10%	143.37	0.9629	0.8712
20%	224.49	0.9766	0.9187
40%	330.33	0.9868	0.9500
60%	345.85	0.9905	0.9700
80%	400.30	0.9942	0.9812
100% (300 checkpoints)	439.77	1.0000	1.0000

Table 3: Diamonds chunk-contribution entropy (mean ± standard deviation over three runs) of FOIF and TracIn under different numbers of chunks (n) and Top-k values.

Method	$K \backslash n$	2	4	6	8	10
FOIF	10	0.9507 ± 0.0619	0.9232 ± 0.0623	0.8948 ± 0.0447	0.8599 ± 0.0996	0.7792 ± 0.0348
	50	0.9729 ± 0.0159	0.9929 ± 0.0067	0.9652 ± 0.0196	0.9631 ± 0.0251	0.9359 ± 0.0174
	100	0.9888 ± 0.0070	0.9951 ± 0.0038	0.9866 ± 0.0098	0.9752 ± 0.0154	0.9747 ± 0.0106
	500	0.9990 ± 0.0005	0.9996 ± 0.0004	0.9988 ± 0.0008	0.9960 ± 0.0011	0.9961 ± 0.0016
TracIn	10	0.8581 ± 0.1261	0.8631 ± 0.1612	0.8530 ± 0.0278	0.7987 ± 0.0145	0.8194 ± 0.0602
	50	0.9946 ± 0.0047	0.9835 ± 0.0207	0.9788 ± 0.0159	0.9645 ± 0.0254	0.9743 ± 0.0043
	100	0.9918 ± 0.0073	0.9901 ± 0.0028	0.9884 ± 0.0045	0.9761 ± 0.0095	0.9852 ± 0.0090
	500	0.9967 ± 0.0031	0.9977 ± 0.0018	0.9978 ± 0.0024	0.9964 ± 0.0018	0.9957 ± 0.0033

Full Feature Space (Padding + MMD). Lower-dimensional datasets are first standardised using the statistics of the 52-feature dataset and then embedded into the common 52-dimensional feature space by padding the missing feature dimensions with zeros, where a zero corresponds to the mean value of the corresponding feature after standardisation. MMD is subsequently computed in this common feature space, measuring the overall distributional change introduced by progressively adding noise features.

Informative Feature Space (Informative Features Only + MMD). MMD is computed using only the original informative features shared by all datasets. Since these informative features remain unchanged throughout feature expansion, this representation serves as a control experiment to verify that the underlying informative distribution is preserved. The MMD between each pair is expected to be always 0.

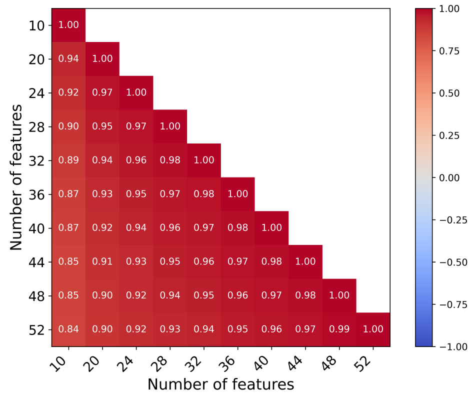
Shared Latent Space (Shared PCA + MMD). All datasets are first represented in the common 52-dimensional feature space and then projected into a low-dimensional latent representation using a single Principal Component Analysis (PCA) model [3] fitted on the 52-feature dataset. By projecting every dataset into the same number of feature dimensions, MMD measures how the dominant underlying data structure evolves as additional noise features are introduced, independent of differences in the original feature dimensionality.

Figure 2 presents the MMD measured under the three dataset representations. As expected, the full feature space exhibits a monotonic increase in MMD as additional noise features are introduced, indicating that the complete feature-space distribution gradually diverges from the original 10-feature dataset. In contrast, the informative feature space consistently yields an MMD of zero across all feature settings, confirming that the original informative features remain unchanged throughout the feature expansion process. Finally, the PCAed experiment also demonstrates an increasing trend, but with substantially smaller MMD values than the full feature space. This suggests that while the added noise features progressively alter the complete feature distribution, the dominant feature structure of the data remains comparatively stable. Overall, these results verify that the observed degradation in influence stability is not caused by changes to the informative data distribution, but rather occurs as non-informative feature dimensions are progressively introduced.

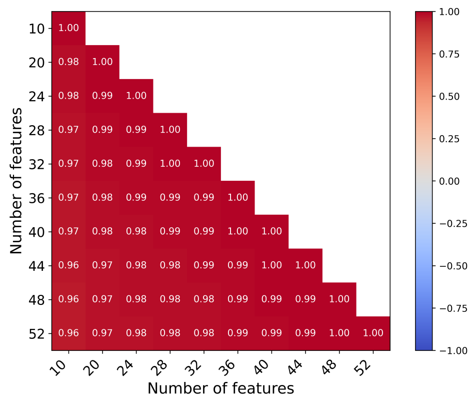
3.2 Zero-Padding Control for Input Dimensionality

To further isolate the effect of feature dimensionality, we replace all added noise features with zero padding while keeping the remaining experimental settings unchanged. This removes the influence of randomly generated noise values and allows us to examine whether increasing the input dimensionality alone affects influence stability.

Figure 3 shows that both FOIF and TracIn achieve higher overall Weighted Kendall Tau values under zero padding than when random noise features are added. Nevertheless, both methods continue to exhibit a gradual decline in ranking stability as the number of feature dimensions increases. This suggests that the observed degradation is not solely caused by the randomness of the injected noise, but is also associated with the increasing input dimensionality itself.



(a) FOIF



(b) TracIn

Figure 3: Effect of feature dimensionality on ranking stability for (a) FOIF and (b) TracIn on SYN_A. Higher values indicate more stable rankings across feature expansions. 0 Padding

4 ADDITIONAL RESULTS FOR HIGH-LOSS SAMPLES

4.1 Results on High-Loss Samples

Figure 4 presents the relationship between training loss and aggregated influence scores for the full ADULT dataset. Compared with the DIAMONDS dataset presented in the main paper, the relationship is noticeably weaker, with substantially greater variance in the influence scores. Nevertheless, both FOIF and TracIn still exhibit a negative regression trend, indicating that training samples with higher loss tend to receive larger-magnitude influence scores.

The weaker correlation is likely attributable to the lower predictive performance of the model on ADULT, resulting in a noisier influence distribution. Despite this increased variability, the overall trend remains consistent with our observations on the synthetic datasets and DIAMONDS, suggesting that high-loss training samples are generally assigned greater influence by both methods.

4.2 Loss-Decile Analysis

To further explain the non-linear relationship between training loss and influence discussed in the main paper, Tables 4 and 5 report a loss-decile analysis for $SYN_A$and the full ADULT dataset, respectively. Training samples are partitioned into ten equal-sized groups according to their training loss, and statistics are computed within each decile.

Table 4: Loss-decile analysis for SYN_A under FOIF. Training samples are divided into ten equal-sized groups according to training loss, where Decile 1 contains the lowest-loss samples and Decile 10 contains the highest-loss samples. The table reports the mean loss, mean signed influence score, fraction of samples with positive influence, and Spearman correlation between loss and signed influence within each decile. The results show a non-linear pattern: influence increases across lower and intermediate loss deciles, but reverses in the highest-loss decile, where most samples receive negative influence.

Loss Decile	Mean Loss	Mean I	Positive Fraction	$\rho_s(loss, I)$
1	0.001012	0.001124	1.000	0.952
2	0.003479	0.003128	1.000	0.777
3	0.007301	0.005542	1.000	0.680
4	0.013394	0.008521	1.000	0.613
5	0.023150	0.012344	1.000	0.455
6	0.039829	0.017452	1.000	0.472
7	0.071284	0.024053	1.000	0.374
8	0.139540	0.030579	0.999	0.112
9	0.341566	0.027195	0.966	-0.381
10	1.524341	-0.132396	0.161	-0.934

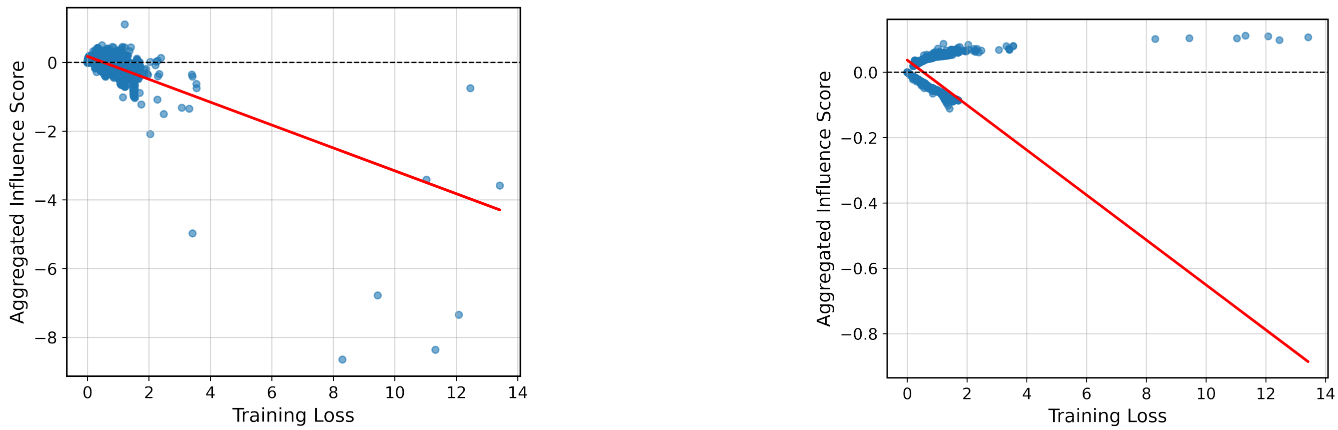
Across both datasets, the mean influence generally increases over the lower and intermediate loss deciles before reversing in the highest-loss deciles, where the average signed influence becomes negative and the fraction of positively influential samples decreases substantially. This explains why a positive overall correlation between loss and signed influence can coexist with a concentration of high-loss samples in the negative-influence tail, as observed for FOIF in the main paper. The same qualitative behaviour is also observed on ADULT , although the trend is weaker due to the greater variability of the dataset.

Table 6 further reports conditional Spearman correlations computed within each class label and within the positive- and negative-influence subsets. These statistics provide a more detailed characterisation of the loss–influence relationship and further illustrate that the association between training loss and signed influence depends on both the dataset and the subset of samples being considered, complementing the overall analysis presented in the main paper.

5 ADDITIONAL RESULT FOR NEIGHBOURHOOD PROXIMITY AND LOCAL DENSITY

5.1 Neighbourhood Distance on a Real Dataset

Figure 5 presents the neighbourhood-distance analysis on the DIAMONDS dataset using the same experimental protocol as in the synthetic experiment described in the main paper. Similar to the observations on SYN_A, the average influence-score difference increases as the neighbourhood distance becomes larger for both FOIF



(a) Adult FOIF

(b) Adult TracIn

Figure 4: The Loss vs Influence Plot with regression line for the FOIF and TracIn for the full size Adult dataset.

Table 5: Loss-decile analysis for Adult under FOIF. Training samples are divided into ten equal-sized groups according to training loss, where Decile 1 contains the lowest-loss samples and Decile 10 contains the highest-loss samples. The table reports the mean loss, mean signed influence score, fraction of samples with positive influence, and Spearman correlation between loss and signed influence within each decile. The results show a non-linear pattern: influence increases across lower and intermediate loss deciles, but reverses in the highest-loss decile, where most samples receive negative influence.

Loss Decile	Mean Loss	Mean I	Positive Fraction	$\rho_s(loss, I)$
1	0.151955	0.020680	0.9296	-0.1404
2	0.235931	0.064526	0.8363	0.9603
3	0.247388	0.146886	0.9991	-0.2718
4	0.250810	0.120953	1.0000	0.1252
5	0.253624	0.108420	0.9998	-0.1688
6	0.258014	0.093810	0.9998	0.4474
7	0.272382	0.107132	0.9975	-0.3059
8	0.404298	0.052191	0.7690	-0.0559
9	1.313176	-0.230780	0.1263	-0.5750
10	1.563272	-0.360498	0.0474	0.5177

Table 6: Conditional Spearman correlations between training loss and influence scores. $\rho_s(L, I \mid y = 0)$ and $\rho_s(L, I \mid y = 1)$ are computed within each class label. $\rho_s(L, |I| \mid I < 0)$ is computed among samples with negative influence scores. $\rho_s(L, I \mid I > 0)$ is computed among samples with positive influence scores.

Dataset	Method	$\rho_s(L, I \mid y=0)$	$\rho_s(L, I \mid y=1)$	$\rho_s(L, \|I\| \mid I<0)$	$\rho_s(L, I \mid I>0)$
SYN_A	FOIF	0.462	0.412	0.933	0.888
SYN_A	TracIn	0.978	-0.991	0.991	0.978
Adult	FOIF	0.175	-0.504	0.457	0.242
Adult	TracIn	0.739	-0.828	0.828	0.739
Diamonds	FOIF	0.758	0.885	0.920	0.971
Diamonds	TracIn	0.999	-0.991	0.991	0.999

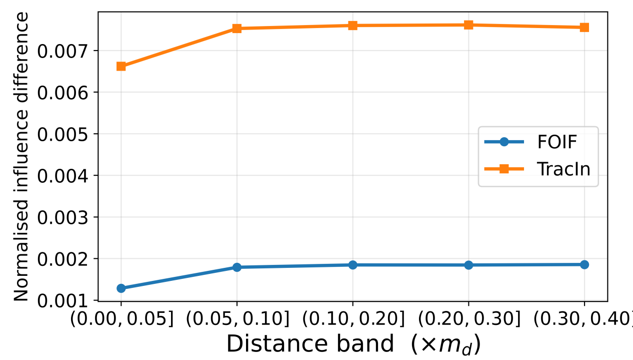


Figure 5: Normalised mean absolute influence-score difference across distance bands for FOIF and TracIn on Diamonds . Distance bands are expressed as multiples of the maximum pairwise distance m_d.

and TracIn. This indicates that nearby training samples tend to receive more similar influence scores, while samples that are farther apart in the feature space exhibit increasingly different influence estimates.

Compared with the synthetic dataset, the overall influence differences on Diamonds are considerably smaller, particularly for FOIF. Nevertheless, the same monotonic trend is preserved, suggesting that the relationship between neighbourhood proximity and influence similarity generalises from the synthetic setting to a real-world dataset.

5.2 Loss-Adjusted Analysis under Additional Density Settings

The SparseDense beta value section We repeat the loss-adjusted analysis under the more extreme density settings $(\sigma_{\text{sparse}}, \sigma_{\text{dense}}) = (2.0, 0.05)$ and $(3.0, 0.01)$. For FOIF, the mean point estimates follow the same qualitative pattern as under $(1.0, 0.1)$: the signed loss slope is positive for dense samples but negative for sparse samples, while influence magnitude increases with loss more strongly in dense regions. However, the variability of the individual coefficients increases substantially as the density contrast becomes more extreme. For TracIn, the derived sparse slopes remain close to zero, and no consistent loss–density relationship is observed.

These results suggest that the FOIF pattern may extend to stronger density contrasts.

6 ADDITIONAL RESULT FOR CLASS IMBALANCE

6.1 Minority Lift under Moderate Imbalance

Table 8 shows that the sign-aware pattern observed under 1:9 imbalance also holds under the 2:8 and 3:7 settings. Lift@Top reaches its ratio-specific maximum for both methods and datasets, indicating that all samples in the positive top-k list belong to the minority class. Lift@Bottom remains substantially lower, particularly for TracIn. Minority samples are also over-represented among the largest influence magnitudes, although this effect becomes weaker on SYN_A as the class distribution becomes more balanced.

6.2 Class-specific Loss-Decile Analysis

To further understand why minority samples are preferentially ranked among the most influential samples, we analyse the loss distribution separately for each class. Tables 9 - 16 report class-specific loss-decile statistics for both FOIF and TracIn under the 1:9 and 3:7 imbalance settings with $SYN_A$and Diamonds datasets.

6.3 Loss-Adjusted Regression Analysis

The loss-decile analysis suggests that both class imbalance and training loss contribute to the observed influence rankings. To disentangle these effects, we fit the loss-adjusted regression model described in the main paper.

Tables 17 and 18 report the regression coefficients for $SYN_A$and Diamonds . Consistent with the main paper, FOIF exhibits a clear interaction between training loss and class membership, whereas the corresponding coefficients for TracIn remain small and less consistent across datasets and imbalance ratios. These additional results further support the conclusion that FOIF is more sensitive to the combined effects of class imbalance and sample difficulty.

7 ADDITIONAL RESULTS FOR DOWNSTREAM TASKS

7.1 Downstream Pruning at a Higher Pruning Rate

The main paper reports the downstream pruning performance when removing 10% of the training samples. To evaluate whether the observed behaviour remains consistent under a more aggressive pruning strategy, Table 19 reports the corresponding results when 20% of the training samples are removed. For completeness, the pruning rate achieving the best performance for each method is also reported in parentheses.

7.2 Additional Results on Standard Datasets

The main paper presents representative downstream noisy detection and pruning results on the standard datasets to illustrate the overall trends. For completeness, this subsection reports the full noisy-label detection and downstream pruning results for SYN_A, Diamonds , and Adult . These additional results are consistent with the observations reported in the main paper and demonstrate that the same conclusions hold across all evaluated datasets.

7.3 Additional Results for Feature Expansion

Tables 26 and 27 report the downstream noisy detection and pruning results for the 52-feature setting.

7.4 Additional Results for Class Imbalance

Tables 28 - 31 report the downstream noisy detection and pruning results under the 1:9 and 3:7 fair-flipping settings.

7.5 Additional Results for Sparse and Dense Subgroups

Tables 32 - 41 report the downstream noisy detection and pruning results under progressively more challenging sparse–dense settings.

REFERENCES

[1] Samyadeep Basu, Philip Pope, and Soheil Feizi. 2021. Influence Functions in Deep Learning Are Fragile. arXiv:2006.14651 [cs.LG] https://arxiv.org/abs/2006.14651

[2] Arthur Gretton, Karsten M. Borgwardt, Malte J. Rasch, Bernhard Schölkopf, and Alexander Smola. 2012. A Kernel Two-Sample Test. *Journal of Machine Learning Research* 13, 25 (2012), 723–773. http://jmlr.org/papers/v13/gretton12a.html

[3] H. Hotelling. 1932. Analysis of a complex of statistical variables into principal components. *Journal of Educational Psychology* 24 (11 1932), 417–441. doi:10.1037/h0071325

Table 7: Loss-adjusted regression results under additional density settings. Values are means and standard deviations across five runs. The sparse slope is $\beta_L + \beta_{LS}$.

Setting	Method	Response	β_L	β_S	β_{LS}	Sparse slope
$(2, 0.05)$	FOIF	Signed	1.2007 ± 1.4836	0.1295 ± 0.0474	-1.6500 ± 1.5025	-0.4493 ± 0.0315
	FOIF	Magnitude	1.1684 ± 1.4472	0.0032 ± 0.0395	-0.8868 ± 1.4406	0.2816 ± 0.0120
	TracIn	Signed	0.0042 ± 0.4277	0.0020 ± 0.0127	-0.0161 ± 0.4260	-0.0118 ± 0.0025
	TracIn	Magnitude	-0.0218 ± 0.3749	-0.0010 ± 0.0118	0.0307 ± 0.3739	0.0089 ± 0.0019
$(3, 0.01)$	FOIF	Signed	1.1394 ± 4.9788	0.2041 ± 0.2177	-1.6491 ± 4.9870	-0.5097 ± 0.0198
	FOIF	Magnitude	1.1165 ± 4.8745	0.0259 ± 0.2095	-0.8338 ± 4.8696	0.2827 ± 0.0076
	TracIn	Signed	0.9629 ± 2.3408	0.0514 ± 0.1025	-0.9922 ± 2.3375	-0.0293 ± 0.0067
	TracIn	Magnitude	0.8012 ± 2.2455	0.0360 ± 0.0995	-0.7791 ± 2.2462	0.0221 ± 0.0032

Table 8: Sign-aware minority lift under moderate class imbalance ($k = 200$, corresponding to 2.5% of 8,000 samples).

Ratio	Dataset	Method	Lift@Top	Lift@Bottom	Lift@AbsTop
2:8	SYN_A	FOIF	5.000	1.600	2.525
		TracIn	5.000	0.000	2.425
	Diamonds	FOIF	5.000	2.375	4.675
		TracIn	5.000	0.450	4.475
3:7	SYN_A	FOIF	3.333	1.267	1.533
		TracIn	3.333	0.000	1.083
	Diamonds	FOIF	3.333	1.917	3.050
		TracIn	3.333	0.533	2.900

Table 9: Class-specific loss-decile statistics under 3:7 imbalance on SYN_A using FOIF. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	621.2 ± 122.65	0.000322 ± 0.000182	0.000227 ± 0.000131	0.999078 ± 0.002061
1	1	179.0 ± 122.65	0.000302 ± 0.000169	0.000352 ± 0.000297	1.000000 ± 0.000000
2	0	644.8 ± 54.22	0.001099 ± 0.000498	0.000644 ± 0.000298	0.999342 ± 0.001471
2	1	155.0 ± 53.98	0.001100 ± 0.000487	0.000987 ± 0.000622	0.998958 ± 0.002329
3	0	646.2 ± 32.77	0.002180 ± 0.000854	0.001117 ± 0.000518	0.999691 ± 0.000690
3	1	153.8 ± 32.77	0.002210 ± 0.000851	0.001731 ± 0.001011	1.000000 ± 0.000000
4	0	622.2 ± 27.95	0.003707 ± 0.001304	0.001723 ± 0.000923	0.999691 ± 0.000691
4	1	177.8 ± 27.95	0.003696 ± 0.001290	0.002501 ± 0.001403	0.999083 ± 0.002051
5	0	609.0 ± 36.15	0.005884 ± 0.001944	0.002473 ± 0.001432	1.000000 ± 0.000000
5	1	191.0 ± 36.15	0.005914 ± 0.001970	0.003526 ± 0.002031	0.996629 ± 0.007537
6	0	586.0 ± 34.56	0.009194 ± 0.003044	0.003416 ± 0.002111	0.999681 ± 0.000714
6	1	214.0 ± 34.56	0.009242 ± 0.003130	0.004894 ± 0.002720	0.997781 ± 0.003042
7	0	556.2 ± 32.32	0.015050 ± 0.005444	0.004900 ± 0.003175	0.998584 ± 0.001426
7	1	243.8 ± 32.32	0.015331 ± 0.005785	0.006693 ± 0.003711	0.999107 ± 0.001996
8	0	497.6 ± 51.70	0.027299 ± 0.011096	0.007445 ± 0.004619	0.996323 ± 0.007029
8	1	302.4 ± 51.70	0.027893 ± 0.011483	0.009508 ± 0.005133	0.991225 ± 0.007549
9	0	440.8 ± 53.50	0.063928 ± 0.030761	0.011815 ± 0.006565	0.989526 ± 0.016654
9	1	359.2 ± 53.50	0.066228 ± 0.031664	0.015048 ± 0.007592	0.992061 ± 0.009526
10	0	376.0 ± 20.11	0.741959 ± 0.215243	-0.042433 ± 0.022946	0.636638 ± 0.130951
10	1	424.0 ± 20.11	0.696790 ± 0.174049	-0.011678 ± 0.009359	0.794339 ± 0.069611

Table 10: Class-specific loss-decile statistics under 3:7 imbalance on SYN_A using TracIn. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	621.2 ± 122.65	0.000322 ± 0.000182	0.000508 ± 0.000361	0.931299 ± 0.097517
1	1	179.0 ± 122.65	0.000302 ± 0.000169	0.001346 ± 0.002365	0.813423 ± 0.235037
2	0	644.8 ± 54.22	0.001099 ± 0.000498	0.000465 ± 0.000464	0.846390 ± 0.221819
2	1	155.0 ± 53.98	0.001100 ± 0.000487	0.001436 ± 0.002513	0.773461 ± 0.286431
3	0	646.2 ± 32.77	0.002180 ± 0.000854	0.000329 ± 0.000555	0.782839 ± 0.280810
3	1	153.8 ± 32.77	0.002210 ± 0.000851	0.001686 ± 0.002822	0.775040 ± 0.282901
4	0	622.2 ± 27.95	0.003707 ± 0.001304	0.000116 ± 0.000784	0.702020 ± 0.355363
4	1	177.8 ± 27.95	0.003696 ± 0.001290	0.001851 ± 0.003132	0.774802 ± 0.284257
5	0	609.0 ± 36.15	0.005884 ± 0.001944	−0.000143 ± 0.001094	0.656988 ± 0.397076
5	1	191.0 ± 36.15	0.005914 ± 0.001970	0.001844 ± 0.003656	0.734059 ± 0.311536
6	0	586.0 ± 34.56	0.009194 ± 0.003044	−0.000341 ± 0.001344	0.616436 ± 0.449907
6	1	214.0 ± 34.56	0.009242 ± 0.003130	0.002087 ± 0.003748	0.730048 ± 0.304696
7	0	556.2 ± 32.32	0.015050 ± 0.005444	−0.000662 ± 0.001735	0.575159 ± 0.474667
7	1	243.8 ± 32.32	0.015331 ± 0.005785	0.002048 ± 0.003933	0.713912 ± 0.326439
8	0	497.6 ± 51.70	0.027299 ± 0.011096	−0.001072 ± 0.002391	0.550719 ± 0.478799
8	1	302.4 ± 51.70	0.027893 ± 0.011483	0.002013 ± 0.003733	0.688168 ± 0.339161
9	0	440.8 ± 53.50	0.063928 ± 0.030761	−0.001984 ± 0.003445	0.493921 ± 0.469374
9	1	359.2 ± 53.50	0.066228 ± 0.031664	0.002197 ± 0.004046	0.659015 ± 0.351180
10	0	376.0 ± 20.11	0.741959 ± 0.215243	−0.004461 ± 0.005811	0.358063 ± 0.382564
10	1	424.0 ± 20.11	0.696790 ± 0.174049	0.002161 ± 0.004860	0.581912 ± 0.391100

Table 11: Class-specific loss-decile statistics under 1:9 imbalance on SYN_A using FOIF. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	781.60 ± 33.13	0.000232 ± 0.000143	0.000197 ± 0.000104	0.989318 ± 0.019580
1	1	23.25 ± 36.14	0.000264 ± 0.000098	0.000482 ± 0.000228	1.000000 ± 0.000000
2	0	786.80 ± 13.55	0.000758 ± 0.000358	0.000569 ± 0.000278	0.981376 ± 0.039488
2	1	13.00 ± 13.71	0.000776 ± 0.000334	0.001046 ± 0.000538	1.000000 ± 0.000000
3	0	787.40 ± 9.71	0.001441 ± 0.000571	0.001013 ± 0.000479	0.984901 ± 0.031634
3	1	12.60 ± 9.71	0.001490 ± 0.000583	0.001887 ± 0.000880	1.000000 ± 0.000000
4	0	783.00 ± 11.13	0.002337 ± 0.000797	0.001528 ± 0.000732	0.984514 ± 0.031808
4	1	17.20 ± 10.96	0.002328 ± 0.000821	0.002711 ± 0.001512	1.000000 ± 0.000000
5	0	784.20 ± 7.82	0.003618 ± 0.001127	0.002220 ± 0.001114	0.986306 ± 0.026521
5	1	15.60 ± 8.08	0.003652 ± 0.001023	0.003826 ± 0.001484	1.000000 ± 0.000000
6	0	774.60 ± 7.50	0.005589 ± 0.001708	0.003022 ± 0.001359	0.989205 ± 0.022017
6	1	25.40 ± 7.50	0.005587 ± 0.001678	0.005164 ± 0.001885	1.000000 ± 0.000000
7	0	769.00 ± 5.24	0.008860 ± 0.002916	0.004098 ± 0.001689	0.985930 ± 0.025066
7	1	31.00 ± 5.24	0.008929 ± 0.002966	0.007796 ± 0.002598	1.000000 ± 0.000000
8	0	733.60 ± 3.85	0.016217 ± 0.006425	0.005615 ± 0.001654	0.972219 ± 0.035675
8	1	66.40 ± 3.85	0.017014 ± 0.006681	0.012321 ± 0.004084	1.000000 ± 0.000000
9	0	622.20 ± 51.16	0.041956 ± 0.020147	0.008403 ± 0.003102	0.951510 ± 0.021978
9	1	177.80 ± 51.16	0.049158 ± 0.023981	0.024372 ± 0.008577	0.998033 ± 0.002695
10	0	377.60 ± 27.55	0.513768 ± 0.127786	−0.051800 ± 0.013248	0.491988 ± 0.198836
10	1	422.40 ± 27.55	0.633796 ± 0.220758	0.017463 ± 0.012105	0.885419 ± 0.052804

Table 12: Class-specific loss-decile statistics under 1:9 imbalance on SYN_A using TracIn. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	781.60 ± 33.13	0.000232 ± 0.000143	−0.000080 ± 0.000339	0.604425 ± 0.317726
1	1	23.25 ± 36.14	0.000264 ± 0.000098	0.006587 ± 0.007623	0.916667 ± 0.166667
2	0	786.80 ± 13.55	0.000758 ± 0.000358	−0.000572 ± 0.000765	0.430046 ± 0.395864
2	1	13.00 ± 13.71	0.000776 ± 0.000334	0.021128 ± 0.027556	0.911111 ± 0.198762
3	0	787.40 ± 9.71	0.001441 ± 0.000571	−0.000932 ± 0.001178	0.371156 ± 0.415153
3	1	12.60 ± 9.71	0.001490 ± 0.000583	0.021497 ± 0.029825	0.893333 ± 0.238514
4	0	783.00 ± 11.13	0.002337 ± 0.000797	−0.001472 ± 0.001926	0.349260 ± 0.415790
4	1	17.20 ± 10.96	0.002328 ± 0.000821	0.022083 ± 0.031375	0.896552 ± 0.231317
5	0	784.20 ± 7.82	0.003618 ± 0.001127	−0.002085 ± 0.002881	0.316100 ± 0.412868
5	1	15.60 ± 8.08	0.003652 ± 0.001023	0.022993 ± 0.030347	0.861538 ± 0.309609
6	0	774.60 ± 7.50	0.005589 ± 0.001708	−0.002900 ± 0.004159	0.303467 ± 0.412665
6	1	25.40 ± 7.50	0.005587 ± 0.001678	0.023016 ± 0.029554	0.905882 ± 0.210453
7	0	769.00 ± 5.24	0.008860 ± 0.002916	−0.003979 ± 0.005800	0.281237 ± 0.415502
7	1	31.00 ± 5.24	0.008929 ± 0.002966	0.026162 ± 0.032072	0.944000 ± 0.125220
8	0	733.60 ± 3.85	0.016217 ± 0.006425	−0.005463 ± 0.007989	0.249050 ± 0.422163
8	1	66.40 ± 3.85	0.017014 ± 0.006681	0.026407 ± 0.033743	0.874286 ± 0.281106
9	0	622.20 ± 51.16	0.041956 ± 0.020147	−0.008357 ± 0.011662	0.220214 ± 0.435206
9	1	177.80 ± 51.16	0.049158 ± 0.023981	0.028822 ± 0.036114	0.893902 ± 0.237241
10	0	377.60 ± 27.55	0.513768 ± 0.127786	−0.019950 ± 0.022396	0.197604 ± 0.418947
10	1	422.40 ± 27.55	0.633796 ± 0.220758	0.030130 ± 0.038241	0.851144 ± 0.331587

Table 13: Class-specific loss-decile statistics under 1:9 imbalance on DIAMONDS using FOIF. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	727.6 ± 6.54	0.000035 ± 0.000002	0.000013 ± 0.000002	0.995874 ± 0.002751
1	1	73.8 ± 5.54	0.000022 ± 0.000001	0.000024 ± 0.000009	1.000000 ± 0.000000
2	0	774.2 ± 9.47	0.000080 ± 0.000004	0.000027 ± 0.000004	0.990198 ± 0.006342
2	1	25.4 ± 7.83	0.000079 ± 0.000005	0.000080 ± 0.000032	1.000000 ± 0.000000
3	0	771.6 ± 3.85	0.000141 ± 0.000008	0.000039 ± 0.000006	0.976671 ± 0.008422
3	1	27.4 ± 3.58	0.000141 ± 0.000008	0.000138 ± 0.000060	1.000000 ± 0.000000
4	0	768.8 ± 4.60	0.000251 ± 0.000016	0.000057 ± 0.000007	0.956315 ± 0.032600
4	1	31.4 ± 4.39	0.000256 ± 0.000013	0.000237 ± 0.000092	1.000000 ± 0.000000
5	0	762.2 ± 3.42	0.000485 ± 0.000027	0.000107 ± 0.000007	0.992667 ± 0.012194
5	1	37.6 ± 3.36	0.000500 ± 0.000030	0.000432 ± 0.000167	1.000000 ± 0.000000
6	0	750.4 ± 3.91	0.001074 ± 0.000057	0.000267 ± 0.000023	1.000000 ± 0.000000
6	1	49.6 ± 3.91	0.001088 ± 0.000050	0.000887 ± 0.000319	1.000000 ± 0.000000
7	0	728.6 ± 8.17	0.002784 ± 0.000116	0.000727 ± 0.000139	1.000000 ± 0.000000
7	1	71.4 ± 8.17	0.002827 ± 0.000214	0.002191 ± 0.000783	1.000000 ± 0.000000
8	0	693.4 ± 7.23	0.007306 ± 0.000196	0.002027 ± 0.000473	0.999710 ± 0.000648
8	1	106.6 ± 7.23	0.007592 ± 0.000129	0.004951 ± 0.001242	1.000000 ± 0.000000
9	0	674.6 ± 12.36	0.027144 ± 0.001632	0.003312 ± 0.000750	0.985673 ± 0.007973
9	1	125.4 ± 12.36	0.026976 ± 0.001635	0.012328 ± 0.002425	1.000000 ± 0.000000
10	0	548.6 ± 11.26	0.192106 ± 0.013011	−0.002445 ± 0.000921	0.671344 ± 0.043340
10	1	251.4 ± 11.26	0.645223 ± 0.019566	0.023947 ± 0.004373	0.911212 ± 0.018305

Table 14: Class-specific loss-decile statistics under 1:9 imbalance on DIAMONDS using TracIn. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	727.6 ± 6.54	0.000035 ± 0.000002	1.867631 ± 0.041623	1.000000 ± 0.000000
1	1	73.8 ± 5.54	0.000022 ± 0.000001	−0.008866 ± 0.002158	0.144485 ± 0.062942
2	0	774.2 ± 9.47	0.000080 ± 0.000004	1.861336 ± 0.042801	1.000000 ± 0.000000
2	1	25.4 ± 7.83	0.000079 ± 0.000005	−0.007036 ± 0.002010	0.137320 ± 0.106261
3	0	771.6 ± 3.85	0.000141 ± 0.000008	1.859663 ± 0.043015	1.000000 ± 0.000000
3	1	27.4 ± 3.58	0.000141 ± 0.000008	−0.005189 ± 0.002084	0.230946 ± 0.106524
4	0	768.8 ± 4.60	0.000251 ± 0.000016	1.861733 ± 0.043578	1.000000 ± 0.000000
4	1	31.4 ± 4.39	0.000256 ± 0.000013	−0.006802 ± 0.003830	0.182608 ± 0.141813
5	0	762.2 ± 3.42	0.000485 ± 0.000027	1.854320 ± 0.042700	1.000000 ± 0.000000
5	1	37.6 ± 3.36	0.000500 ± 0.000030	−0.007069 ± 0.001856	0.141458 ± 0.121288
6	0	750.4 ± 3.91	0.001074 ± 0.000057	1.846716 ± 0.042373	1.000000 ± 0.000000
6	1	49.6 ± 3.91	0.001088 ± 0.000050	−0.006824 ± 0.002430	0.155440 ± 0.055132
7	0	728.6 ± 8.17	0.002784 ± 0.000116	1.825163 ± 0.038311	1.000000 ± 0.000000
7	1	71.4 ± 8.17	0.002827 ± 0.000214	−0.008424 ± 0.003424	0.126753 ± 0.044781
8	0	693.4 ± 7.23	0.007306 ± 0.000196	1.817954 ± 0.037861	1.000000 ± 0.000000
8	1	106.6 ± 7.23	0.007592 ± 0.000129	−0.011943 ± 0.003715	0.037349 ± 0.014601
9	0	674.6 ± 12.36	0.027144 ± 0.001632	1.809495 ± 0.038104	1.000000 ± 0.000000
9	1	125.4 ± 12.36	0.026976 ± 0.001635	−0.015712 ± 0.003465	0.025867 ± 0.007993
10	0	548.6 ± 11.26	0.192106 ± 0.013011	1.796679 ± 0.036969	1.000000 ± 0.000000
10	1	251.4 ± 11.26	0.645223 ± 0.019566	−0.022062 ± 0.002546	0.034685 ± 0.018293

Table 15: Class-specific loss-decile statistics under 3:7 imbalance on DIAMONDS using FOIF. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	378.0 ± 74.35	0.000017 ± 0.000013	0.000006 ± 0.000008	1.000000 ± 0.000000
1	1	422.4 ± 73.32	0.000007 ± 0.000004	0.000001 ± 0.000005	0.999635 ± 0.000816
2	0	664.4 ± 8.62	0.000049 ± 0.000034	0.000015 ± 0.000017	1.000000 ± 0.000000
2	1	134.8 ± 9.12	0.000047 ± 0.000030	0.000005 ± 0.000004	1.000000 ± 0.000000
3	0	692.8 ± 7.85	0.000107 ± 0.000066	0.000029 ± 0.000031	1.000000 ± 0.000000
3	1	106.0 ± 9.97	0.000107 ± 0.000064	0.000010 ± 0.000009	1.000000 ± 0.000000
4	0	663.2 ± 7.26	0.000222 ± 0.000116	0.000052 ± 0.000049	1.000000 ± 0.000000
4	1	136.4 ± 7.77	0.000230 ± 0.000123	0.000024 ± 0.000020	1.000000 ± 0.000000
5	0	629.0 ± 8.19	0.000498 ± 0.000190	0.000098 ± 0.000076	1.000000 ± 0.000000
5	1	171.4 ± 7.96	0.000521 ± 0.000201	0.000052 ± 0.000042	1.000000 ± 0.000000
6	0	533.4 ± 12.58	0.001255 ± 0.000330	0.000216 ± 0.000133	1.000000 ± 0.000000
6	1	266.2 ± 12.28	0.001295 ± 0.000324	0.000127 ± 0.000101	1.000000 ± 0.000000
7	0	528.0 ± 25.27	0.003531 ± 0.000629	0.000493 ± 0.000226	1.000000 ± 0.000000
7	1	272.2 ± 25.28	0.003366 ± 0.000602	0.000335 ± 0.000250	1.000000 ± 0.000000
8	0	530.0 ± 20.60	0.010093 ± 0.001223	0.001276 ± 0.000407	1.000000 ± 0.000000
8	1	269.8 ± 20.36	0.010244 ± 0.001222	0.001113 ± 0.000588	1.000000 ± 0.000000
9	0	521.2 ± 21.66	0.046819 ± 0.004076	0.002750 ± 0.000531	0.997290 ± 0.002242
9	1	278.8 ± 21.66	0.047508 ± 0.004472	0.004276 ± 0.001365	1.000000 ± 0.000000
10	0	460.0 ± 37.65	0.453033 ± 0.036042	−0.02552 ± 0.000947	0.632644 ± 0.026654
10	1	340.0 ± 37.65	0.624414 ± 0.081076	0.002669 ± 0.001975	0.818555 ± 0.007999

Table 16: Class-specific loss-decile statistics under 3:7 imbalance on Diamonds using TracIn. The table reports sample count, mean loss, mean signed influence, and the fraction of samples with positive influence.

Loss Decile	Label	Count	Mean Loss	Mean Influence	Positive Fraction
1	0	378.0 ± 74.36	0.000017 ± 0.000013	1.871184 ± 0.056050	1.000000 ± 0.000000
1	1	424.4 ± 73.32	0.000007 ± 0.000004	−0.007746 ± 0.001132	0.000000 ± 0.000000
2	0	664.4 ± 8.62	0.000049 ± 0.000034	1.869941 ± 0.054202	1.000000 ± 0.000000
2	1	134.8 ± 9.12	0.000047 ± 0.000030	−0.007417 ± 0.001972	0.000000 ± 0.000000
3	0	692.8 ± 7.85	0.000107 ± 0.000066	1.866917 ± 0.055889	1.000000 ± 0.000000
3	1	106.0 ± 9.97	0.000107 ± 0.000064	−0.007384 ± 0.001354	0.000000 ± 0.000000
4	0	663.2 ± 7.26	0.000222 ± 0.000116	1.866595 ± 0.057343	1.000000 ± 0.000000
4	1	136.4 ± 7.77	0.000230 ± 0.000123	−0.008022 ± 0.001187	0.000000 ± 0.000000
5	0	629.0 ± 8.19	0.000498 ± 0.000190	1.862772 ± 0.057273	1.000000 ± 0.000000
5	1	171.4 ± 7.96	0.000521 ± 0.000201	−0.009390 ± 0.001329	0.000000 ± 0.000000
6	0	533.4 ± 12.58	0.001255 ± 0.000330	1.854157 ± 0.055360	1.000000 ± 0.000000
6	1	266.2 ± 12.28	0.001295 ± 0.000324	−0.011932 ± 0.001817	0.000000 ± 0.000000
7	0	528.0 ± 25.27	0.003531 ± 0.000629	1.839270 ± 0.050404	1.000000 ± 0.000000
7	1	272.2 ± 25.28	0.003366 ± 0.000602	−0.013931 ± 0.001132	0.000000 ± 0.000000
8	0	530.0 ± 20.60	0.010093 ± 0.001223	1.830474 ± 0.050548	1.000000 ± 0.000000
8	1	269.8 ± 20.36	0.010244 ± 0.001222	−0.017071 ± 0.003169	0.000000 ± 0.000000
9	0	521.2 ± 21.66	0.046819 ± 0.004076	1.828871 ± 0.042116	1.000000 ± 0.000000
9	1	278.8 ± 21.66	0.047508 ± 0.004472	−0.018662 ± 0.002049	0.000000 ± 0.000000
10	0	460.0 ± 37.65	0.453033 ± 0.036042	1.842047 ± 0.042398	1.000000 ± 0.000000
10	1	340.0 ± 37.65	0.624414 ± 0.081076	−0.019614 ± 0.001879	0.000000 ± 0.000000

Table 17: Loss-adjusted regression results under class imbalance on SYN_A. Values are means and standard deviations across five runs. The majority loss slope is β_L, and the minority loss slope is $\beta_L + \beta_{LM}$.

Setting	Method	Response	β_L	β_M	β_{LM}	Minority slope
1:9	FOIF	Signed	−0.316603 ± 0.060252	0.037549 ± 0.017957	0.197969 ± 0.092459	−0.118633 ± 0.044927
	FOIF	Magnitude	0.236879 ± 0.031481	0.012507 ± 0.007174	−0.136736 ± 0.034445	0.100143 ± 0.022668
	TracIn	Signed	−0.042411 ± 0.039803	0.030456 ± 0.038985	0.044765 ± 0.045089	0.002353 ± 0.006318
	TracIn	Magnitude	0.040648 ± 0.037827	0.024771 ± 0.029985	−0.037542 ± 0.034726	0.003106 ± 0.005286
3:7	FOIF	Signed	−0.206236 ± 0.042639	0.007532 ± 0.004004	0.064702 ± 0.024620	−0.141533 ± 0.036928
	FOIF	Magnitude	0.168327 ± 0.029398	0.001973 ± 0.001293	−0.043933 ± 0.015422	0.124395 ± 0.022254
	TracIn	Signed	−0.007203 ± 0.006629	0.002179 ± 0.004749	0.007119 ± 0.008804	−0.000084 ± 0.002445
	TracIn	Magnitude	0.006454 ± 0.006596	0.001693 ± 0.002226	−0.004485 ± 0.006303	0.001969 ± 0.000827

Table 18: Loss-adjusted regression results under class imbalance on Diamonds . Values are means and standard deviations across five runs. The majority loss slope is β_L, and the minority loss slope is $\beta_L + \beta_{LM}$.

Setting	Method	Response	β_L	β_M	β_{LM}	Minority slope
1:9	FOIF	Signed	−0.061603 ± 0.021353	0.008578 ± 0.001408	0.064260 ± 0.021530	0.002656 ± 0.004174
	FOIF	Magnitude	0.063703 ± 0.020318	0.007911 ± 0.001857	−0.036902 ± 0.018194	0.026801 ± 0.005502
	TracIn	Signed	−0.208177 ± 0.029933	−1.857260 ± 0.043584	0.193678 ± 0.033459	−0.014499 ± 0.004508
	TracIn	Magnitude	−0.073816 ± 0.009961	−1.033458 ± 0.012451	0.087954 ± 0.007127	0.014138 ± 0.004139
3:7	FOIF	Signed	−0.021206 ± 0.006976	0.001304 ± 0.001612	0.003621 ± 0.022481	−0.017585 ± 0.027766
	FOIF	Magnitude	0.022422 ± 0.006538	0.000240 ± 0.000818	−0.000441 ± 0.014210	0.021981 ± 0.019220
	TracIn	Signed	−0.036989 ± 0.073659	−1.867925 ± 0.054144	0.024272 ± 0.072892	−0.012716 ± 0.001678
	TracIn	Magnitude	−0.012930 ± 0.025556	−1.037006 ± 0.017478	0.025371 ± 0.026415	0.012442 ± 0.001674

Table 19: Downstream pruning utility after removing suspicious training samples and retraining from scratch. For each method, we report the test accuracy obtained by pruning 20%, of the training data. The pruning rate that gives the best result is shown in parentheses.

Setting	No Pruning	Training Loss	FOIF	TracIn
SYN_A, 20% random flip	0.7620	**0.7840**	0.7640	0.7540
Diamonds , 20% random flip	**0.9680**	0.9600	0.9660	0.9040
SYN_A, 1:9 fair flip	0.8960	0.9160	**0.9520**	0.8700
SYN_A, 3:7 fair flip	0.9460	**0.9600**	0.9560	0.9180
SYN_A, 5:5	0.9320	**0.9520**	0.9440	0.9260
SYN_B, sparse 1 dense 0.1	0.9400	0.9440	0.9120	0.9280
SYN_B, sparse 2 dense 0.05	0.8200	0.8160	0.8200	0.8100

Table 20: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Harder Setting: Sep 3

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1919	0.1600	0.0800	0.1850	0.1850	0.1925	0.2888
Training Loss	0.6011	0.7050	0.3525	0.5656	0.5656	0.4712	0.7069
FOIF	0.5665	0.6600	0.3300	0.5488	0.5488	0.4633	0.6950
TracIn	0.3744	0.5450	0.2725	0.3912	0.3912	0.2988	0.4481

Table 21: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Harder Setting: Sep 3

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.6694	0.7620	0
FOIF (10%)	0.7550	0.7840	528
FOIF (20%)	0.8428	0.7640	878
FOIF (30%)	0.9427	0.7760	1112
TracIn (10%)	0.7476	0.7780	436
TracIn (20%)	0.8261	0.7540	626
TracIn (30%)	0.8798	0.6940	717
Training Loss (10%)	0.7511	0.7900	564
Training Loss (20%)	0.8481	0.7840	905
Training Loss (30%)	0.9561	0.7800	1131
Random (10%)	0.6667	0.7840	128
Random (20%)	0.6752	0.7740	296
Random (30%)	0.6739	0.7600	462

Table 22: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Diamonds Dataset

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1977	0.1903	0.0952	0.1955	0.1955	0.1984	0.2975
Training Loss	0.9795	0.9991	0.4995	0.9220	0.9220	0.6632	0.9948
FOIF	0.9636	0.9951	0.4976	0.9162	0.9162	0.6466	0.9699
TracIn	0.6065	0.9173	0.4586	0.4976	0.4976	0.3322	0.4983

Table 23: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Diamonds Dataset

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.7830	0.9680	0
FOIF (10%)	0.8708	0.9660	5318
FOIF (20%)	0.9727	0.9660	9792
FOIF (30%)	0.9923	0.9660	10366
TracIn (10%)	0.8644	0.9040	4902
TracIn (20%)	0.8989	0.9040	5318
TracIn (30%)	0.9166	0.7820	5326
Training Loss (10%)	0.8709	0.9760	5339
Training Loss (20%)	0.9758	0.9600	9854
Training Loss (30%)	0.9999	0.9680	10632
Random (10%)	0.7811	0.9520	1017
Random (20%)	0.7795	0.9560	2089
Random (30%)	0.7792	0.9600	3180

Table 24: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Adult Dataset

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.2025	0.2035	0.1017	0.2035	0.2035	0.2048	0.3072
Training Loss	0.4705	0.5018	0.2509	0.4935	0.4935	0.4884	0.7327
FOIF	0.4256	0.4714	0.2357	0.4732	0.4732	0.4755	0.7132
TracIn	0.4154	0.4662	0.2331	0.4739	0.4739	0.4811	0.7216

Table 25: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Adult Dataset

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.6746	0.8120	0
FOIF (10%)	0.7505	0.8080	2108
FOIF (20%)	0.8374	0.8080	4232
FOIF (30%)	0.9533	0.8000	6379
TracIn (10%)	0.7444	0.8060	2085
TracIn (20%)	0.8376	0.8000	4239
TracIn (30%)	0.9562	0.8000	6454
Training Loss (10%)	0.7485	0.8080	2244
Training Loss (20%)	0.8390	0.7980	4414
Training Loss (30%)	0.9631	0.8100	6553
Random (10%)	0.6754	0.8060	910
Random (20%)	0.6722	0.8100	1820
Random (30%)	0.6742	0.8080	2748

Table 26: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Feature 52

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1919	0.1600	0.0800	0.1850	0.1850	0.1925	0.2888
Training Loss	0.7376	0.8275	0.4138	0.6769	0.6769	0.5438	0.8156
FOIF	0.7082	0.8062	0.4031	0.6688	0.6688	0.5350	0.8025
TracIn	0.5971	0.7150	0.3575	0.5744	0.5744	0.4579	0.6869

Table 27: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Feature 52

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.7314	0.8280	0
FOIF (10%)	0.8462	0.8340	645
FOIF (20%)	0.9436	0.8380	1070
FOIF (30%)	0.9914	0.8380	1284
TracIn (10%)	0.8211	0.8260	572
TracIn (20%)	0.8983	0.8040	919
TracIn (30%)	0.9748	0.7720	1099
Training Loss (10%)	0.8461	0.8260	662
Training Loss (20%)	0.9484	0.8440	1083
Training Loss (30%)	0.9955	0.8340	1305
Random (10%)	0.7279	0.8240	128
Random (20%)	0.7231	0.8020	296
Random (30%)	0.7255	0.8100	462

Table 28: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Class Imbalance 1:9 Fair Flipping

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.2000	0.2100	0.1050	0.2000	0.2000	0.2058	0.3088
Training Loss	0.9573	0.9788	0.4894	0.9319	0.9319	0.6612	0.9919
FOIF	0.7209	0.8188	0.4094	0.7606	0.7606	0.5779	0.8669
TracIn	0.2288	0.1812	0.0906	0.1812	0.1812	0.2375	0.3562

Table 29: **Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Class Imbalance 1:9 Fair Flipping**

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.7821	0.8960	0
FOIF (10%)	0.8574	0.9440	655
FOIF (20%)	0.9477	0.9520	1217
FOIF (30%)	0.9871	0.9440	1387
TracIn (10%)	0.8062	0.9300	145
TracIn (20%)	0.8308	0.8700	290
TracIn (30%)	0.8795	0.8300	570
Training Loss (10%)	0.8712	0.9180	783
Training Loss (20%)	0.9833	0.9160	1491
Training Loss (30%)	0.9998	0.8820	1587
Random (10%)	0.7829	0.8920	168
Random (20%)	0.7800	0.8900	320
Random (30%)	0.7812	0.8940	494

Table 30: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Class Imbalance 3:7 Fair Flipping

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.2050	0.2125	0.1062	0.2169	0.2169	0.2042	0.3062
Training Loss	0.9511	0.9725	0.4862	0.9194	0.9194	0.6571	0.9856
FOIF	0.8735	0.9325	0.4662	0.8538	0.8538	0.6179	0.9269
TracIn	0.6005	0.7325	0.3662	0.5950	0.5950	0.4629	0.6944

Table 31: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Class Imbalance 3:7 Fair Flipping

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.7768	0.9460	0
FOIF (10%)	0.8651	0.9560	746
FOIF (20%)	0.9673	0.9560	1366
FOIF (30%)	0.9923	0.9400	1483
TracIn (10%)	0.8507	0.9380	586
TracIn (20%)	0.9092	0.9180	952
TracIn (30%)	0.9541	0.8760	1111
Training Loss (10%)	0.8664	0.9560	778
Training Loss (20%)	0.9781	0.9600	1471
Training Loss (30%)	0.9998	0.9300	1577
Random (10%)	0.7771	0.9420	170
Random (20%)	0.7792	0.9500	347
Random (30%)	0.7782	0.9500	490

Table 32: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Sparse 1 Dense 0.1 Half sparse Half Dense in both labels

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1919	0.1600	0.0800	0.1850	0.1850	0.1925	0.2888
Training Loss	0.9671	0.9875	0.4938	0.9112	0.9112	0.6604	0.9906
FOIF	0.8862	0.9575	0.4788	0.8619	0.8619	0.6179	0.9269
TracIn	0.9248	0.9775	0.4888	0.8750	0.8750	0.6279	0.9419

Table 33: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Sparse 1 Dense 0.1 Half sparse Half Dense in both labels

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.7768	0.9640	0
FOIF (10%)	0.8622	0.9620	766
FOIF (20%)	0.9636	0.9580	1379
FOIF (30%)	0.9855	0.9540	1483
TracIn (10%)	0.8664	0.9540	782
TracIn (20%)	0.9620	0.9500	1400
TracIn (30%)	0.9991	0.9440	1507
Training Loss (10%)	0.8651	0.9580	790
Training Loss (20%)	0.9762	0.9620	1458
Training Loss (30%)	1.0000	0.9640	1585
Random (10%)	0.7753	0.9600	128
Random (20%)	0.7741	0.9580	296
Random (30%)	0.7777	0.9520	462

Table 34: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Sparse 2 Dense 0.05 Half sparse Half Dense in both labels

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1919	0.1600	0.0800	0.1850	0.1850	0.1925	0.2888
Training Loss	0.8178	0.9375	0.4688	0.7625	0.7625	0.5900	0.8850
FOIF	0.7424	0.8625	0.4312	0.7419	0.7419	0.5738	0.8606
TracIn	0.7193	0.7812	0.3906	0.7012	0.7012	0.5733	0.8600

Table 35: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Sparse 2 Dense 0.05 Half sparse Half Dense in both labels

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.7347	0.8920	0
FOIF (10%)	0.8174	0.8860	690
FOIF (20%)	0.9187	0.9040	1187
TracIn (10%)	0.8190	0.8820	625
TracIn (20%)	0.9122	0.8620	1122
Training Loss (10%)	0.8175	0.8920	750
Training Loss (20%)	0.9234	0.9020	1220
Random (10%)	0.7354	0.8980	128
Random (20%)	0.7364	0.8720	296

Table 36: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Sparse 3 Dense 0.01 Half sparse Half Dense in both labels

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1919	0.1600	0.0800	0.1850	0.1850	0.1925	0.2888
Training Loss	0.7616	0.9475	0.4738	0.6925	0.6925	0.5417	0.8125
FOIF	0.6025	0.7188	0.3594	0.6894	0.6894	0.5400	0.8100
TracIn	0.6251	0.6975	0.3488	0.6888	0.6888	0.5417	0.8125

Table 37: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Sparse 3 Dense 0.01 Half sparse Half Dense in both labels

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.7151	0.8460	0
FOIF (10%)	0.7986	0.8400	575
FOIF (20%)	0.8942	0.8340	1103
TracIn (10%)	0.7965	0.8380	558
TracIn (20%)	0.8867	0.8280	1102
Training Loss (10%)	0.7944	0.8300	758
Training Loss (20%)	0.9011	0.8460	1108
Random (10%)	0.7129	0.8240	128
Random (20%)	0.7155	0.8200	296

Table 38: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Sparse 5 Dense 0.001 Half sparse Half Dense in both labels

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1919	0.1600	0.0800	0.1850	0.1850	0.1925	0.2888
Training Loss	0.7108	0.9550	0.4775	0.6406	0.6406	0.5092	0.7638
FOIF	0.3421	0.2500	0.1250	0.3875	0.3875	0.4933	0.7400
TracIn	0.4355	0.6612	0.3306	0.4431	0.4431	0.3058	0.4588

Table 39: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Sparse 5 Dense 0.001 Half sparse Half Dense in both labels

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.6898	0.7960	0
FOIF (10%)	0.7533	0.8000	200
FOIF (20%)	0.8325	0.7980	620
TracIn (10%)	0.7639	0.7920	529
TracIn (20%)	0.8389	0.8020	709
Training Loss (10%)	0.7654	0.7700	764
Training Loss (20%)	0.8700	0.7940	1025
Random (10%)	0.6801	0.7840	128
Random (20%)	0.6862	0.8000	296

Table 40: Noisy-label detection performance under 20% randomly flipped training labels. Higher AUPRC, Precision@K and Recall@K indicate better noisy-label detection. Sparse 10 Dense 0.0001 Half sparse Half Dense in both labels

Method	AUPRC	P@10%	R@10%	P@20%	R@20%	P@30%	R@30%
Random	0.1919	0.1600	0.0800	0.1850	0.1850	0.1925	0.2888
Training Loss	0.7174	0.9550	0.4775	0.6138	0.6138	0.4858	0.7288
FOIF	0.2313	0.2162	0.1081	0.2169	0.2169	0.3350	0.5025
TracIn	0.3077	0.6225	0.3113	0.4119	0.4119	0.2946	0.4419

Table 41: Model performance after removing suspicious training samples under 20% randomly flipped training labels. The model is retrained from scratch after pruning. Sparse 10 Dense 0.0001 Half sparse Half Dense in both labels

Method	Training Accuracy	Test Accuracy	# Noisy Samples Removed
No Pruning	0.6666	0.7700	0
FOIF (10%)	0.7376	0.7780	173
FOIF (20%)	0.8023	0.7700	347
TracIn (10%)	0.7486	0.7520	498
TracIn (20%)	0.8280	0.7620	659
Training Loss (10%)	0.7481	0.7520	764
Training Loss (20%)	0.8375	0.7780	982
Random (10%)	0.6733	0.7720	128
Random (20%)	0.6677	0.7760	296